

# Giang Do (James)

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Google Scholar

## RESEARCH INTERESTS

My research focuses on the efficiency, scalability, and reliability of large foundation models. In particular, I work on **Sparse Mixture-of-Experts (MoE)** for training and inference of large language models—studying routing, representational collapse, and the emergence of specialised experts—and on **controllable and robust reasoning** in LLMs. A unifying thread of my PhD is the *binding problem*: how foundation models compose, separate, and route information across modular components.

## EDUCATION

- Deakin University** VIC, Australia  
• *PhD in Information Technology* Jun 2024 – Present  
**Thesis topic:** Improving Foundation Models by Addressing the Binding Problem.  
**Supervisors:** Prof. Truyen Tran, Dr. Hung Le (Applied AI Institute, A<sup>2</sup>I<sup>2</sup>).
- The University of Tennessee at Chattanooga** TN, USA  
• *M.Sc. in Computer Science; GPA: 4.0/4.0 (Class rank: 1st)* Jan 2023 – May 2024  
**Selected coursework:** Mathematical Statistics, Machine Learning, Advanced Systems Software.
- VNU University of Science** Hanoi, Vietnam  
• *M.Sc. in Data Science; GPA: 3.75/4.0 (Class rank: 2nd)* Dec 2020 – Dec 2022  
**Thesis:** Identifying Constructive Comments for Vietnamese News (8.2/9.0; best thesis award).  
**Supervisor:** Dr. Tuan-Anh Hoang, RMIT Vietnam University.
- Hanoi University of Science & Technology (HUST)** Hanoi, Vietnam  
• *B.Tech. in Information Technology; GPA: 3.36/4.0 (Top 7%)* Aug 2019 – Jun 2021  
**Thesis:** Constructive Comment Classification (9.25/10.0; best thesis award).
- Foreign Trade University** Hanoi, Vietnam  
• *B.A. in International Business Economics; Classification: Good* Aug 2010 – Jun 2014

## PUBLICATIONS

(**Bold** = author. \* denotes first author.)

### Refereed Conference & Journal Papers

- Eigenvectors of Experts are Training-free Non-collapsing Routers**  
*Giang Do\**, Hung Le, Truyen Tran  
**ICML 2026 (Spotlight).**
- Rethinking Sparse Mixture of Experts from a Unified Perspective**  
*Giang Do\**, Hung Le, Truyen Tran  
**ICML 2026.**
- Do Domain-specific Experts Exist in MoE-based LLMs?**  
*Giang Do\**, Hung Le, Truyen Tran  
**ACL 2026 (Findings).**
- SimSMoE: Toward Efficient Training of Mixture of Experts via Solving Representational Collapse**  
*Giang Do\**, Hung Le, Truyen Tran  
**NAACL 2025 (Findings).**
- On the Role of Discrete Representation in Sparse Mixture of Experts**  
*Giang Do\**, Kha Pham, Hung Le, Truyen Tran  
**Transactions on Machine Learning Research (TMLR), 2025.**
- HyperRouter: Towards Efficient Training and Inference of Sparse Mixture of Experts**  
*Giang Do\**, Khiem Le, TrungTin Nguyen, Quang Pham, Binh T. Nguyen, Thanh-Nam Doan, Chenghao Liu, Savitha Ramasamy, Xiaoli Li, Steven Hoi  
**EMNLP 2023 (Main Conference).**

### Preprints & Under Review

- Measuring and Improving Robustness of Large Reasoning Models under Task Interruption**  
*Giang Do\**, Tri Minh Nguyen, Hung Le, Truyen Tran  
**Under review, 2026.**
- Think in Episodes: Controlling Reasoning in Language Models via Episodic Inference**  
*Giang Do\**, Hung Le, Truyen Tran  
**Under review, 2026.**

## 9. CompeteSMoE: Effective Training of Sparse Mixture of Experts via Competition

Quang Pham, **Giang Do**, Huy Nguyen, TrungTin Nguyen, Chenghao Liu, Mina Sartipi, Binh T. Nguyen, Savitha Ramasamy, Xiaoli Li, Steven Hoi, Nhat Ho

Under review. [arXiv:2402.02526](#).

## RESEARCH EXPERIENCE

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- **Applied Artificial Intelligence Institute (A<sup>2</sup>I<sup>2</sup>), Deakin University** VIC, Australia  
*Research Assistant* Oct 2024 – Present
  - **Radiology Report Generation Evaluation:** Surveyed existing evaluation methods for radiology report generation and proposed a knowledge-graph-based metric better aligned with expert clinical judgement.
  - **3D Infant Modelling and Video Tracking:** Developed 2D detection and 3D tracking pipelines for infant videos to support downstream behavioural and clinical analysis.
- **Center for Urban Informatics and Progress (CUIP), UT Chattanooga** TN, USA  
*Graduate Research Assistant* Jan 2023 – May 2024
  - **Large Language Models:** Investigated hypernetworks and Sparse Mixture-of-Experts to improve training stability and efficiency of large language models (Transformer-XL, RoBERTa). This work led to the EMNLP 2023 *HyperRouter* paper.
  - **Camera-LiDAR Data Fusion:** Studied state-of-the-art 2D and 3D object detection models for fusing camera and LiDAR data in urban-scene perception.

## INDUSTRY EXPERIENCE

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- **Panasonic R&D Center Vietnam** Hanoi, Vietnam  
*AI Project Leader* Feb 2021 – Jan 2023
  - **Digital Twin for Buildings:** Led a team of 4 engineers to design and deliver a digital-twin system for building modelling and monitoring (3D BIM + Unity).
  - **3D Object Detection:** Trained and fine-tuned state-of-the-art 3D object detectors, achieving +5% accuracy over pre-trained baselines.
  - **Unsupervised Ceiling-Light Recognition:** Proposed an unsupervised point-cloud method that recognised ceiling lights with >90% accuracy.
  - **Semantic 3D Mapping:** Designed an algorithm mapping 2D instance segmentation outputs onto 3D building models.
- **Techcombank (Vietnam Technological & Commercial Joint Stock Bank)** Hanoi, Vietnam  
*Senior Data Scientist* Oct 2019 – Jan 2021
  - **Pricing Models:** Built ML systems to recommend bond/stock buy-sell prices for retail clients.
  - **Anomaly Detection:** Developed models to detect abnormal logins and trading patterns for security and compliance.
- **Toyota Motor Vietnam** Hanoi, Vietnam  
*Data Analyst* Jan 2016 – Jul 2019
  - **Sales Forecasting:** Built time-series forecasting models informing production and marketing decisions for top management.

## HONORS AND AWARDS

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- Spotlight Paper, ICML 2026 (*Eigenvectors of Experts are Training-free Non-collapsing Routers*).
- UTC Outstanding Graduate Student Award, 2023–2024.
- Best Master's Thesis Award, VNU University of Science — “Identifying Constructive Comments for Vietnamese News”, Dec 2022.
- Best Employee Award, Panasonic R&D Vietnam, Jul 2022.
- Best Project Award, Panasonic R&D Vietnam, Dec 2021.
- Best Bachelor Thesis Award, HUST — “Constructive Comment Classification”, Jun 2021.
- Vingroup Science and Technology Scholarship for M.Sc. Data Science students, 2021–2022.

## ACADEMIC SERVICE

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- Reviewer: AISTATS 2026, NeurIPS 2026.

## TECHNICAL SKILLS

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- **Programming:** Python (primary), R, SQL, C/C++, Bash
- **ML / DL Frameworks:** PyTorch, HuggingFace Transformers, DeepSpeed, TensorFlow, scikit-learn
- **Infrastructure:** Linux, Slurm, Docker, Git; multi-GPU training (A100/H100)

## REFERENCES

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### Prof. Truyen Tran

Head of AI, Health and Science  
Applied Artificial Intelligence Institute (A<sup>2</sup>I<sup>2</sup>)  
Deakin University  
✉ [truyen.tran@deakin.edu.au](mailto:truyen.tran@deakin.edu.au)

### Dr. Hung Le

Senior Lecturer  
Applied Artificial Intelligence Institute (A<sup>2</sup>I<sup>2</sup>)  
Deakin University  
✉ [thai.le@deakin.edu.au](mailto:thai.le@deakin.edu.au)